Unit 5, Lesson 9: Congruence in Two-Dimensional Figures – Part 1

Benchmark

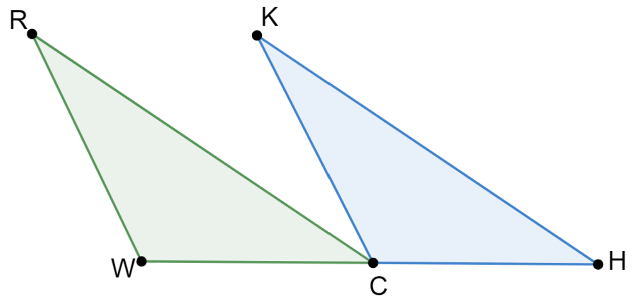
MA.912.GR.1.6 Solve mathematical and real-world problems involving congruence or similarity in two-dimensional figures.

Learning Target

☐ I can solve mathematical problems involving congruence.

5.9.1 Warm-Up: Bell Work

Triangles RCW and KHC are shown.



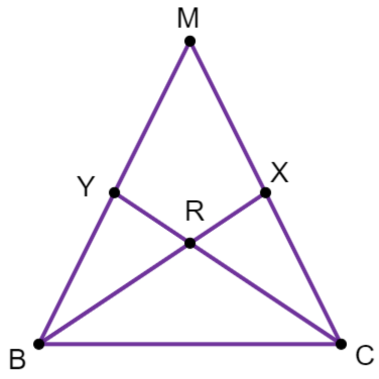
Given: $RW \parallel KC$, C is the midpoint of $\overline{WH}$, $\overline{WR} \cong \overline{CK}$
Prove: $\angle WRC \cong \angle CKH$

1. Complete the missing statements and reasons in the two-column proof.

Statement	Reason
1. C is the midpoint of $\overline{WH}$	1. Given
2.	2. Definition of midpoint
3. $RW \parallel KC$	3. Given
4.	4.
5.	5. Given
6.	6.
7. $\angle WRC \cong \angle CKH$	7.

5.9.2 Collaboration: Using CPCTC to Determine Measures

Triangle MBC is shown where $MX = 3.6$, $MB = 7.2$, $RX = 2$, $BC = 6.5$, $RB = 3.9$, $m\angle YBR = 29°$, $m\angle CXR = 83°$, and $m\angle MCB = 63°$.



Work with your partner to complete the following.

1. How many different triangles are in ΔMBC?

2. If $\Delta BRY \cong \Delta CRX$, then determine the measure of each.

   $RC =$ ________ $m\angle RYB =$ _______

3. If $\Delta BXM \cong \Delta CYM$, then determine the measure of each.

   $CY =$ ________ $CM =$ ________ $m\angle MCY =$ _______

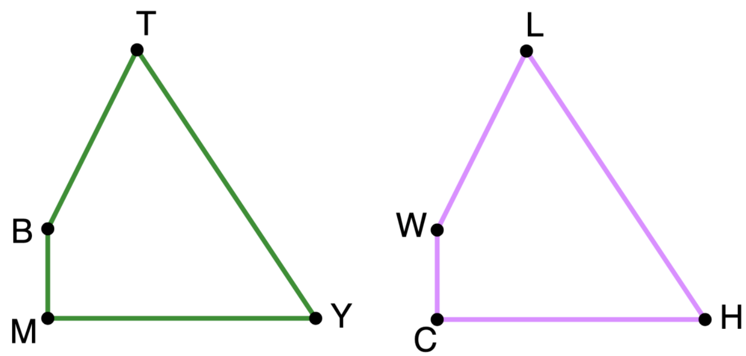
5.9.3 Guided Instruction: Corresponding Parts of Congruent Polygons

Any polygon that can be mapped to another polygon using rigid transformations shows that all of the parts of the polygons are congruent. This understanding can then be applied to find measurements of the two congruent polygons.

Corresponding Parts of Congruent Polygons are Congruent

If two polygons are congruent, then all corresponding parts of the polygons are congruent.

1. Congruent quadrilaterals $TYMB$ and $LHCW$ are shown.



Some measurements of the angles in quadrilaterals $TYMB$ and $LHCW$ are given.

$m\angle YMB = 90°$, $m\angle WLH = 60°$, $m\angle MBT + m\angle CHL = 210°$, $m\angle BTY = (2x - 4)°$, $m\angle MYT = (x + 25)°$

a. Determine each.

$m\angle BTY$

$m\angle CWL$

Some measurements of the sides of quadrilaterals $TYMB$ and $LHCW$ are given.

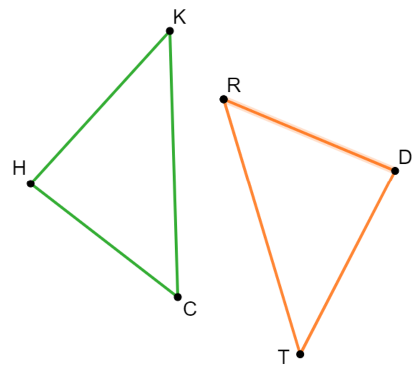
$HC = 5k$, $LW = 4k - 0.5$, $YT = 6k$, and $BM = k + 1$.

b. Determine the value of k if the perimeter of quadrilateral $TYMB$ is 24.5.

c. What is the length of BT?

2. The diagram shows ΔHKC and ΔTDR where $\Delta HKC \cong \Delta DTR$.

Some measurements of the sides are given.
$KH = 5y - 2$,
$DR = 6y - 5$, and
$TD = 4y + 3$



Arthur used the information to determine the length of KH. His work is shown.

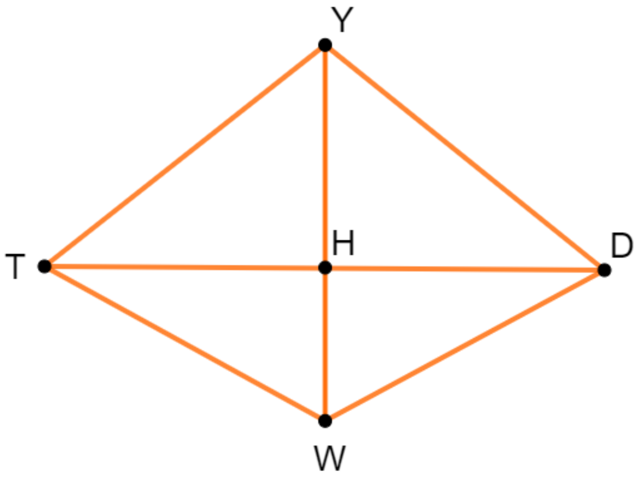
Arthur's Work
$5y - 2 = 6y - 5$ $-2 = y - 5$ $3 = y$
$KH = 5(3) - 2$ $KH = 13$

a. Find Arthur's error.

b. Help Arthur understand and learn from his mistake. Write a brief note to Arthur explaining what his mistake was and how he can learn from the mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

1. Quadrilateral $YDWT$ is shown where $\overline{YH}$ is a perpendicular bisector of $\overline{TD}$.



a. How many different triangles are in Quadrilateral $YDWT$?

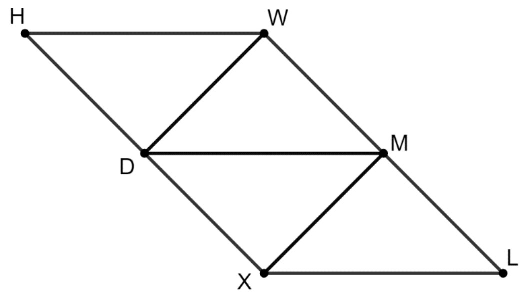
b. If $\Delta DHY \cong \Delta THY$, $YD = \sqrt{130}$, $YH = 7$, and $m\angle HTY = 38°$, determine the following measures.

$HD =$ ________ $m\angle DYH =$ _______ $TH =$ ________

c. If $\Delta DHW \cong \Delta THW$, $WH = 5$, and $m\angle HWD = 61°$, use these and all other known measurements to determine the following measures.

$m\angle TWH =$ ______ $m\angle HTW =$ ______ $TW =$ ______

2. Quadrilateral $HWLX$ is shown.



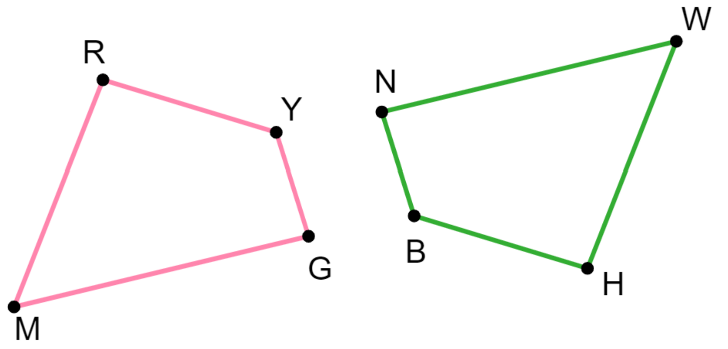
$HWMX \cong LXDW$ and $\Delta XMD \cong \Delta WDM$. The perimeter of $HWMX$ is 40.5.

a. Use the given measurements to find the value of x.
$HW = 2x - 2$, $XD = \frac{3}{2}x - 1$, $HX = 2x + 5$, $WD = 2x - 2$

b. Find the length of MX.

5.9.5 Wrap-Up: Error Analysis

1. Quadrilaterals $RYGM$ and $HBNW$ are shown.



$RYGM \cong HBNW$ and the following information is also true.

$m\angle RYG = (10x - 6)°$, $MG = 2k + 3$, $NW = 4k - 29$,
$m\angle MGY = (7x - 5)°$, $m\angle BNW = 86°$, $HB = k + 9$

Hector used the given information to write the two equations shown.

$$10x - 6 = 86 \qquad\qquad 4k - 29 = 2k + 3$$

One of Hector's equations is NOT correct. Determine which one, and correct his error.

Unit 5, Lesson 10: Congruence in Two-Dimensional Figures – Part 2

Benchmark

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Learning Target

☐ I can solve real-world problems involving congruence.